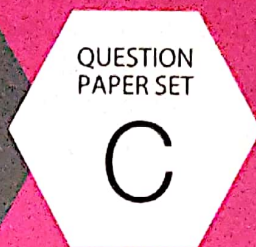


**SOF INTERNATIONAL
MATHEMATICS OLYMPIAD
2023-24**



DO NOT OPEN THIS BOOKLET UNTIL ASKED TO DO SO

Total Questions: 50 | Time: 1 hr.

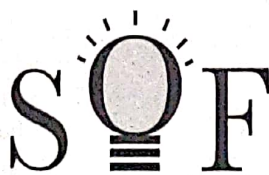
Guidelines for the Candidate

1. You will get additional ten minutes to fill up information about yourself on the OMR Sheet, before the start of the exam.
2. Write your **Name, School Code, Class, Section, Roll No.** and **Mobile Number** clearly on the **OMR Sheet** and do not forget to sign it. We will share your marks / result and other information related to SOF exams on your mobile number.
3. The Question Paper comprises four sections:
Logical Reasoning (15 Questions), **Mathematical Reasoning** (20 Questions), **Everyday Mathematics** (10 Questions) and **Achievers Section** (5 Questions)
Each question in Achievers Section carries 3 marks, whereas all other questions carry one mark each.
4. All questions are compulsory. There is no negative marking. Use of calculator is not permitted.
5. There is only ONE correct answer. Choose only ONE option for an answer.
6. To mark your choice of answers by darkening the circles on the OMR Sheet, use **HB Pencil** or **Blue / Black ball point pen** only. E.g.
Q.16: Rahul bought 4 kg 90 g of apples, 2 kg 60 g of grapes and 5 kg 300 g of mangoes. The total weight of all the fruits he bought is _____.
A. 11.450 kg B. 11.000 kg C. 11.350 kg D. 11.250 kg
As the correct answer is option A, you must darken the circle corresponding to option A on the OMR Sheet.
7. Rough work should be done in the blank space provided in the booklet.
8. Return the OMR Sheet to the invigilator at the end of the exam.
9. Please fill in your personal details in the space provided before attempting the paper.

16. ● (B) (C) (D)

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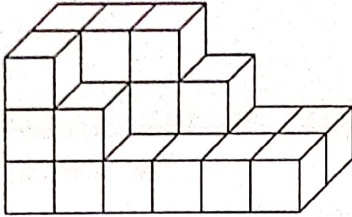
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LOGICAL REASONING

1. Riya starts walking from her college towards North. After walking for 10 m, she turns right and walks 25 m. She then turns right again and walks 50 m. Finally, she turns right and walk for 25 m to reach her home. How far is Riya now from her college?

A. 30 m
B. 40 m
C. 25 m
D. 50 m

2. Count the number of cubes in the given figure.

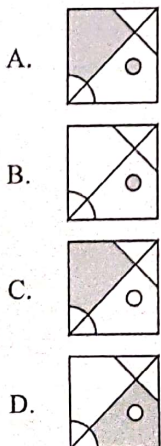
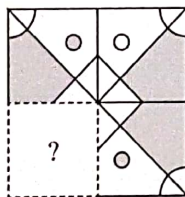


A. 20
B. 21
C. 22
D. 18

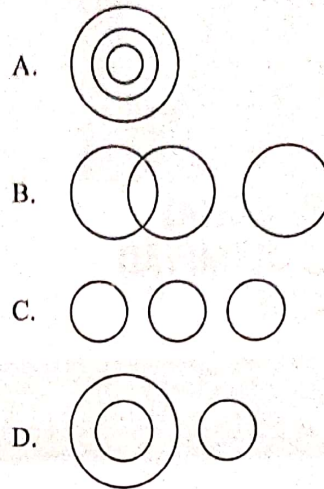
3. In a certain code language, 'EXAMS' is written as '67249' and 'SHARED' is written as '912563'. How will 'ASHRAM' be written in that code language?

A. 291524
B. 791624
C. 295142
D. 921523

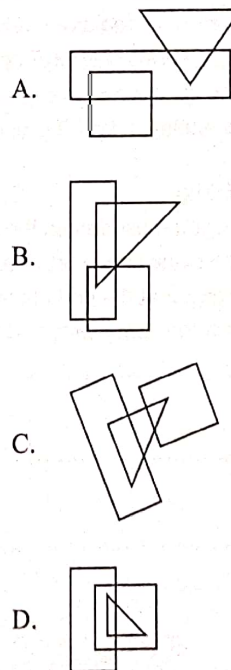
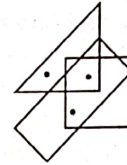
4. Which of the following figures will complete the pattern in the given figure?



5. Which of the following Venn diagrams best represents the relationship amongst, "Moon, Earth and India"?



6. Select a figure from the options which satisfies the same conditions of placement of dots as in the given figure.

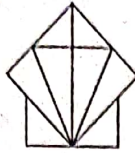


7. In a class of 28 students, Gautam is 15th from the bottom. What will be his rank from the top?

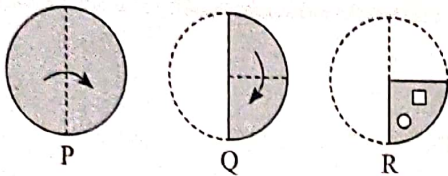
A. 14th
B. 15th
C. 13th
D. 16th

8. Find the minimum number of straight lines required to draw the given figure.

- A. 8
B. 9
C. 10
D. 11



9. A set of three figures P, Q and R showing a sequence of folding of a piece of paper is given. Fig. R shows the manner in which the folded paper has been cut. Select a figure from the options which shows the unfolded form of Fig. R.

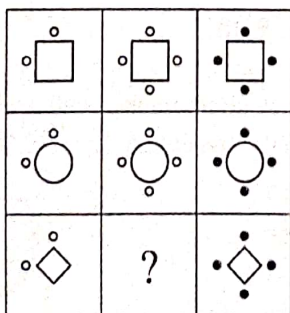


- A.
B.
C.
D.

10. If 'A' denotes '+', 'B' denotes '+', 'C' denotes '-' and 'D' denotes '×', then the value of 17 B 33 A 11 C 5 D 2 =

- A. 30
B. 0
C. 21
D. 10

11. Select a figure from the options which will complete the given figure matrix.



- A.
B.
C.
D.

12. Pointing towards Sara, Vijay said, "She is the daughter of my father's father." How is Sara related to Vijay?

- A. Aunt
B. Mother
C. Sister
D. Granddaughter

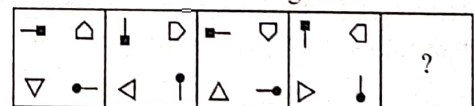
13. Select the correct mirror image of the given combination of letters and symbols, if the mirror is placed vertically to the left.

F@M#L?Y

- A. Y?J#M@F
B. Y?J#M@F
C. Y?J#M@F
D. Y?L#M@F

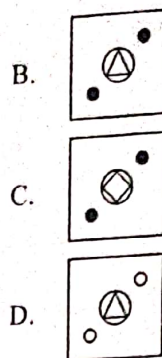
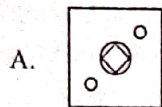
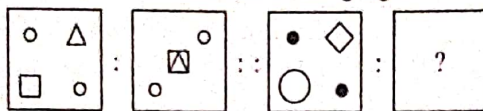
14. Select a figure from the options which will continue the same series as established by the Problem Figures.

Problem Figures



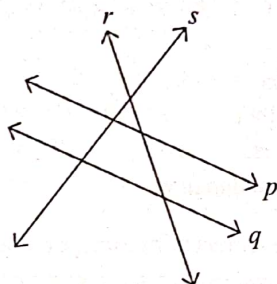
- A.
B.
C.
D.

15. There is a certain relationship between the pair of figures on the either side of ::. Identify the relationship of the left pair and find the missing figure.



MATHEMATICAL REASONING

16. How many pairs of intersecting lines are there in the given figure?

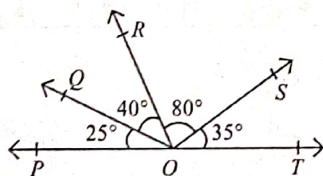


- A. 4
B. 3
C. 5
D. 6

17. Which of the following gives the maximum value?

- A. $-5 + 7 - 17 + 0$
B. $25 - 31 + 15 - 6$
C. $-18 - 37 + 45 + 5$
D. $50 - 45 - 40 + 15$

18. How many obtuse angles are formed in the given figure?



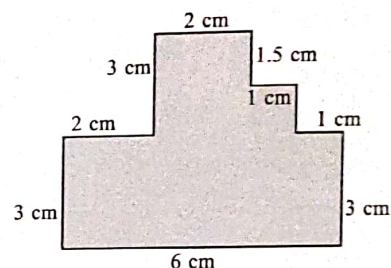
- A. 5
B. 4
C. 3
D. 2

19. How many of the following numbers have at least one line of symmetry?

351067

- A. 1
B. 2
C. 3
D. 4

20. Find the area of the given figure (not drawn to scale).



- A. 25.5 sq. cm
B. 23.5 sq. cm
C. 25 sq. cm
D. 24 sq. cm

21. Which of the following options are arranged in ascending order?

- A. $\frac{1}{2}, \frac{7}{18}, \frac{5}{9}, \frac{7}{27}$
B. $\frac{5}{9}, \frac{1}{2}, \frac{7}{18}, \frac{7}{27}$
C. $\frac{7}{27}, \frac{7}{18}, \frac{1}{2}, \frac{5}{9}$
D. $\frac{7}{27}, \frac{5}{9}, \frac{7}{18}, \frac{1}{2}$

22. Find the smallest number which when diminished by 3 is exactly divisible by 21, 28, 36 and 45.

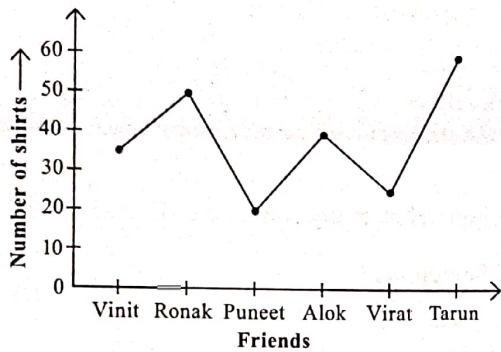
- A. 1263
B. 1560
C. 1143
D. 1290

23. Compare and fill in the box.

$$\text{CCCLXXVI} + \text{CDXIV} \square \text{DCXIX} + \text{CCLXVIII}$$

- A. $>$
 B. $=$
 C. $<$
 D. Can't be determined

Direction (24-25) : The given line graph shows the number of shirts bought by six friends in a particular year. Study the graph carefully and answer the following questions.



24. If Alok's father gives him 12 more shirts, then how many more shirts does he have now than the number of shirts Virat bought?

- A. 22
 B. 32
 C. 27
 D. 35

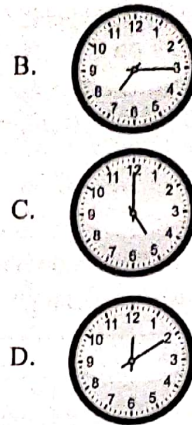
25. Find the ratio of number of shirts bought by Tarun to the number of shirts bought by Vinit and Ronak together.

- A. 17:12
 B. 13:25
 C. 12:13
 D. 12:17

26. Equation for the statement: 'Twice the product of m and n is equal to thrice of their difference' is _____.

- A. $3mn = 2(m - n)$
 B. $mn = 2(m - n)$
 C. $2mn = m - n$
 D. $2mn = 3(m - n)$

27. Which of the following hands of the clocks shows $\frac{1}{4}$ of a revolution?



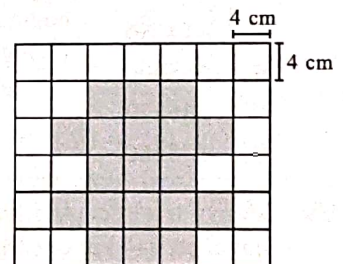
28. Which of the following options equals to $\left(\frac{0.5}{0.05} + \frac{0.05}{0.5}\right)$?

- A. 0.5
 B. 10.1
 C. 1.1
 D. 5.5

29. If the number 455(?)656 is completely divisible by 3, then the smallest whole number in place of (?) will be _____.

- A. 8
 B. 0
 C. 1
 D. 2

30. Find the perimeter of the shaded part of the given figure.



- A. 84 cm
 B. 92 cm
 C. 96 cm
 D. None of these

31. If $a = 35$, $b = 11$ and $c = 23$, then find $a \times (c - b)$.

- A. 375
 B. 495
 C. 235
 D. 420

32. Read the given statements carefully and select the correct option.

- I. Predecessor of largest 7-digit even number is even.
 II. In International system of numeration, 54137083 is written as fifty four million one hundred thirty seven thousand and eighty three.

- A. Only I is true
 B. Only II is true
 C. Both I and II are true
 D. Neither I nor II is true

33. How many of the following curves are open?



- A. 3
C. 2
B. 4
D. 5

34. A car was parked in a parking lot from 5:00 p.m. to 9:45 p.m. How much does it cost for parking?

Parking Rates	
First hour	₹ 18.50
Every additional half an hour or part thereof	₹ 5

- A. ₹ 74
B. ₹ 33.50
C. ₹ 35
D. ₹ 58.50

35. Find the sum of (place value of 9) and (the difference between the place value of 3 and face value of 5) in 4325907.

- A. 300895
B. 11
C. 30895
D. 3800895

EVERYDAY MATHEMATICS

36. An apple costs ₹ x and a mango costs ₹ 5 more than the cost of an apple. Find the total cost (in ₹) of 5 apples and 4 mangoes.

- A. $5x + 20$
B. $5x + 10$
C. $9x + 20$
D. None of these

37. Priyanka needs to type 1950 pages. If the number of words on each page is 315, then find the total number of words she will type in all.

- A. 614250
B. 585000
C. 635270
D. 595480

38. Manya travelled 465 km towards South and Ananya travelled 644 km towards North from the same point. Find the distance between their final destination.

- A. 1109 km
B. -1109 km
C. 1080 km
D. 179 km

39. There were 64 boys and some girls at a party. After one hour, 24 boys left and 30 more girls joined the party. After this, the ratio of the number of boys to the number of girls at the party become 2 : 5. How many girls were there at first?

- A. 65
B. 60
C. 70
D. 130

40. Mrs. Sinha is a piano teacher. She taught a total of 24 hours in the month of September. Each class lasted

$\frac{3}{4}$ hour. How many classes did she teach in the month of September?

- A. 42
B. 40
C. 28
D. 32

41. Three tankers contain 4200 L, 5040 L and 6750 L of water. What is the maximum capacity that can measure all the different quantities exactly?

- A. 42 L
B. 50 L
C. 60 L
D. 30 L

42. In a day of 24 hours, if Garima spends 8 hours in office, 4 hours in travelling, 5 hours in playing with her son and rest time in sleeping, then what fraction of a day she spends in sleeping?

- A. $\frac{26}{7}$
B. $\frac{7}{26}$
C. $\frac{7}{24}$
D. $\frac{24}{7}$

43. Sonali buys 25 pencils and 25 pens for distributing to her friends on her birthday. If the cost of a pencil is ₹ 8 and that of a pen is ₹ 15, then how much total amount does she spend?

- A. ₹ 575
B. ₹ 615
C. ₹ 425
D. ₹ 585

44. Mayank travelled 352.15 km in three days. If he travelled 115.28 km on first day and 79.50 km on second day, then how much distance did he travel on the third day?

- A. 98.51 km

- B. 157.37 km
C. 85.40 km
D. None of these

45. The total cost of flooring a room is ₹ 2160. The rate of flooring is ₹ 45 per square metre. If the room is 8 metres long, then find its breadth.

- A. 11 m
B. 7 m
C. 12 m
D. 6 m

ACHIEVERS SECTION

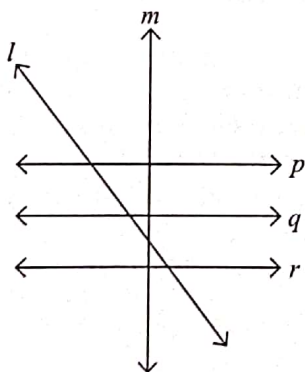
46. Read the given statements carefully and select the correct option.

Statement-I: The value of $(30) + (-53) + (-63) + (-40) + (20) - (-21) - (21) - (-53) + (-30) - (-63) - (-40) - (20)$ is -15 .

Statement-II: On subtracting the greatest 5-digit number from the additive inverse of 1000, we get 100000.

- A. Both Statement-I and Statement-II are true.
B. Both Statement-I and Statement-II are false.
C. Statement-I is true but Statement-II is false.
D. Statement-I is false but Statement-II is true.

47. Study the given figure carefully and answer the following questions.



- (i) How many pairs of perpendicular lines are there in the given figure?
(ii) How many pairs of intersecting lines are there in the given figure?
(iii) How many concurrent lines are there in the given figure?

- | | (i) | (ii) | (iii) |
|----|-----|------|-------|
| A. | 2 | 6 | 2 |
| B. | 3 | 7 | 0 |
| C. | 4 | 5 | 1 |
| D. | 3 | 3 | 2 |

48. Out of 150 students, 45 likes to eat noodles, 30 likes to eat pizza, 25 likes to eat burger and rest of the students likes to eat pasta. Then, find and match the ratio.

Column-I

Column-II

- | | |
|---|----------|
| (i) Number of students who likes to eat pizza to that of pasta is | (p) 5:9 |
| (ii) Number of students who likes to eat noodles to that of total number of students is | (q) 3:5 |
| (iii) Number of students who likes to eat burger to that of noodles is | (r) 3:10 |

- A. (i) → (q); (ii) → (r); (iii) → (p)
B. (i) → (p); (ii) → (r); (iii) → (q)
C. (i) → (p); (ii) → (q); (iii) → (r)
D. (i) → (q); (ii) → (p); (iii) → (r)

49. Read the given statements carefully and state T for true and F for false.

- (i) The length and breadth of a rectangular floor are in the ratio 7 : 5. If the perimeter of the floor is 216 m, then its area is 2835 sq. m.
(ii) A table top measures 4 m 25 cm by 3 m 10 cm. The cost of lacing the boundary of table top, if the cost of 1 m lace is ₹ 15, is ₹ 110.
(iii) The perimeter of a regular pentagon of side 5 m is 20 m.

- | | (i) | (ii) | (iii) |
|----|-----|------|-------|
| A. | F | T | F |
| B. | F | F | T |
| C. | T | F | F |
| D. | T | F | T |

50. Fill in the blanks and select the correct option.

- (i) The predecessor of largest 6-digit number is _____.
- (ii) Seventy million fifty thousand two hundred seventy nine is written as _____.
- (iii) The smallest 6-digit number that can be formed

by using the digits 2, 3, 1 and 7 (using each digit at least once) is _____.

	(i)	(ii)	(iii)
A.	999998	70,050,279	111237
B.	999997	70,050,279	111237
C.	999999	70,500,279	111237
D.	999998	7,050,279	122237

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